

Table 2. Hazard Ratios for All-Cause Mortality
Survey-weighted Cox Proportional Hazards Model, NHANES 1999-2000

The SURVEYPHREG Procedure

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Model Information		
Data Set	PROJ.ANALYTIC	
Dependent Variable	fu_months	
Censoring Variable	died_allcause	
Censoring Value(s)	0	
Weight Variable	WTMEC2YR	Full Sample 2 Year MEC Exam Weight
Stratum Variable	SDMVSTRA	Masked Variance Pseudo-Stratum
Cluster Variable	SDMVPSU	Masked Variance Pseudo-PSU
Ties Handling	BRESLOW	

Number of Observations Read	2188
Number of Observations Used	2188
Sum of Weights Read	78174935
Sum of Weights Used	78174935

Design Summary	
Number of Strata	13
Number of Clusters	27

Class Level Information		
Class	Levels	Values
smoke_status	3	Current Former Never
bmi_cat	4	Normal Obese Overweight Underweight
diabetes	2	0 1

Summary of the Number of Event and Censored Values			
Total	Event	Censored	Percent Censored
2188	1150	1038	47.44

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*Summary of the Weighted Number of Event
and Censored Values*

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Total	Event	Censored	Percent Censored
78174935	33344347	44830588	57.35

Variance Estimation

Method Taylor Series

Model Fit Statistics

Criterion	Without Covariates	With Covariates
-2 LOG L	1.1933E9	1.1826E9
AIC	1.1933E9	1.1826E9

Testing Global Null Hypothesis: BETA=0

Test	Test Statistic	Num DF	Den DF	p-Value
Likelihood Ratio (Unadj.)	10701788.6	9	Infty	<.0001
Likelihood Ratio (Adj.)	111.2021	4.541	Infty	<.0001
Wald	98.7425	9	6	<.0001

Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	t Value	Pr > t	Hazard Ratio
sbp_avg	14	0.022779	0.001372	16.61	<.0001	1.023
LBXTC	14	-0.000517	0.000752	-0.69	0.5027	0.999
LBDHDL	14	-0.000531	0.003725	-0.14	0.8886	0.999
smoke_status Current	14	0.527245	0.081704	6.45	<.0001	1.694
smoke_status Former	14	0.164575	0.068981	2.39	0.0317	1.179
bmi_cat Obese	14	-0.247610	0.140905	-1.76	0.1007	0.781
bmi_cat Overweight	14	-0.201742	0.078501	-2.57	0.0222	0.817
bmi_cat Underweight	14	0.654230	0.298881	2.19	0.0460	1.924
diabetes 1	14	0.687761	0.104650	6.57	<.0001	1.989

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Systolic BP (per 10 mmHg): Hazard Ratios
for sbp_avg

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Description	Point Estimate	95% Wald Confidence Limits	
sbp_avg Unit=10	1.256	1.219	1.293

Total cholesterol (per 40 mg/dL): Hazard Ratios for LBXTC

Description	Point Estimate	95% Wald Confidence Limits	
LBXTC Unit=40	0.980	0.918	1.045

Hazard Ratios for smoke_status

Description	Point Estimate	95% Wald Confidence Limits	
smoke_status Current vs Former	1.437	1.146	1.803
smoke_status Current vs Never	1.694	1.422	2.019
smoke_status Former vs Never	1.179	1.017	1.367

Hazard Ratios for bmi_cat

Description	Point Estimate	95% Wald Confidence Limits	
bmi_cat Normal vs Obese	1.281	0.947	1.733
bmi_cat Normal vs Overweight	1.224	1.034	1.448
bmi_cat Normal vs Underweight	0.520	0.274	0.987
bmi_cat Obese vs Overweight	0.955	0.741	1.232
bmi_cat Obese vs Underweight	0.406	0.224	0.734
bmi_cat Overweight vs Underweight	0.425	0.224	0.806

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Hazard Ratios for diabetes			
Description	Point Estimate	95% Wald Confidence Limits	
diabetes 0 vs 1	0.503	0.402	0.629

Model Information		
Data Set	PROJ.ANALYTIC	
Dependent Variable	fu_months	
Censoring Variable	died_allcause	
Censoring Value(s)	0	
Weight Variable	WTMEC2YR	Full Sample 2 Year MEC Exam Weight
Stratum Variable	SDMVSTRA	Masked Variance Pseudo-Stratum
Cluster Variable	SDMVPSU	Masked Variance Pseudo-PSU
Ties Handling	BRESLOW	

Number of Observations Read	2188
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Sum of Weights Read	78174935
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Design Summary	
Number of Strata	13
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Class Level Information		
Class	Levels	Values
smoke_status	3	Current Former Never
bmi_cat	4	Normal Obese Overweight Underweight
diabetes	2	0 1
RIAGENDR	2	1 2
RIDRETH1	5	1 2 3 4 5

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*Summary of the Number of Event
and Censored Values*

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			Percent
Total	Event	Censored	Censored
2188	1150	1038	47.44

*Summary of the Weighted Number of Event
and Censored Values*

			Percent
Total	Event	Censored	Censored
78174935	33344347	44830588	57.35

Variance Estimation

Method	Taylor Series
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Model Fit Statistics

Criterion	Without Covariates	With Covariates
-2 LOG L	1.1933E9	1.15298E9
AIC	1.1933E9	1.15298E9

Testing Global Null Hypothesis: BETA=0

Test	Test Statistic	Num DF	Den DF	p-Value
Likelihood Ratio (Unadj.)	40315960.6	16	Infty	<.0001
Likelihood Ratio (Adj.)	410.0502	6.591	Infty	<.0001
Wald	.	16	.	.

Warning: The model has more parameters than there are independent variance units. Variance estimates may be unstable.

Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	t Value	Pr > t	Hazard Ratio
sbp_avg	14	0.006037	0.001395	4.33	0.0007	1.006
LBXTC	14	0.000301	0.000892	0.34	0.7409	1.000
LBDHDL	14	0.002515	0.003778	0.67	0.5164	1.003
smoke_status Current	14	0.922265	0.092619	9.96	<.0001	2.515
smoke_status Former	14	0.025697	0.076446	0.34	0.7417	1.026

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Analysis of Maximum Likelihood Estimates

<i>Parameter</i>	<i>DF</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>t Value</i>	<i>Pr > t </i>	<i>Hazard Ratio</i>
<i>bmi_cat Obese</i>	14	0.119667	0.105324	1.14	0.2750	1.127
<i>bmi_cat Overweight</i>	14	-0.179047	0.083993	-2.13	0.0512	0.836
<i>bmi_cat Underweight</i>	14	0.580315	0.191355	3.03	0.0090	1.787
<i>diabetes 1</i>	14	0.499609	0.109078	4.58	0.0004	1.648
<i>RIDAGEYR</i>	14	0.096106	0.003851	24.96	<.0001	1.101
<i>RIAGENDR 2</i>	14	-0.498627	0.096206	-5.18	0.0001	0.607
<i>RIDRETH1 1</i>	14	-0.232902	0.164116	-1.42	0.1777	0.792
<i>RIDRETH1 2</i>	14	-0.523971	0.120861	-4.34	0.0007	0.592
<i>RIDRETH1 4</i>	14	0.058673	0.123623	0.47	0.6424	1.060
<i>RIDRETH1 5</i>	14	0.088916	0.207000	0.43	0.6741	1.093
<i>DMDEDUC2</i>	14	-0.144454	0.029230	-4.94	0.0002	0.865

*Systolic BP (per 10 mmHg): Hazard Ratios
for sbp_avg*

<i>Description</i>	<i>Point Estimate</i>	<i>95% Wald Confidence Limits</i>	
<i>sbp_avg Unit=10</i>	1.062	1.031	1.094

Total cholesterol (per 40 mg/dL): Hazard Ratios for LBXTC

<i>Description</i>	<i>Point Estimate</i>	<i>95% Wald Confidence Limits</i>	
<i>LBXTC Unit=40</i>	1.012	0.938	1.093

Hazard Ratios for smoke_status

<i>Description</i>	<i>Point Estimate</i>	<i>95% Wald Confidence Limits</i>	
<i>smoke_status Current vs Former</i>	2.451	1.908	3.149
<i>smoke_status Current vs Never</i>	2.515	2.062	3.068
<i>smoke_status Former vs Never</i>	1.026	0.871	1.209

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Hazard Ratios for bmi_cat

<i>Description</i>	<i>Point Estimate</i>	<i>95% Wald Confidence Limits</i>	
<i>bmi_cat Normal vs Obese</i>	0.887	0.708	1.112
<i>bmi_cat Normal vs Overweight</i>	1.196	0.999	1.432
<i>bmi_cat Normal vs Underweight</i>	0.560	0.371	0.844
<i>bmi_cat Obese vs Overweight</i>	1.348	1.067	1.703
<i>bmi_cat Obese vs Underweight</i>	0.631	0.386	1.030
<i>bmi_cat Overweight vs Underweight</i>	0.468	0.306	0.716

Hazard Ratios for diabetes

<i>Description</i>	<i>Point Estimate</i>	<i>95% Wald Confidence Limits</i>	
<i>diabetes 0 vs 1</i>	0.607	0.480	0.767

Model Information

<i>Data Set</i>	PROJ.ANALYTIC	
<i>Dependent Variable</i>	fu_months	
<i>Censoring Variable</i>	died_cvd	
<i>Censoring Value(s)</i>	0	
<i>Weight Variable</i>	WTMEC2YR	Full Sample 2 Year MEC Exam Weight
<i>Stratum Variable</i>	SDMVSTRA	Masked Variance Pseudo-Stratum
<i>Cluster Variable</i>	SDMVPSU	Masked Variance Pseudo-PSU
<i>Ties Handling</i>	BRESLOW	

<i>Number of Observations Read</i>	2188
<i>Number of Observations Used</i>	2188
<i>Sum of Weights Read</i>	78174935
<i>Sum of Weights Used</i>	78174935

Design Summary

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Class	Levels	Values
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bmi_cat	4	Normal Obese Overweight Underweight
diabetes	2	0 1
RIAGENDR	2	1 2
RIDRETH1	5	1 2 3 4 5

*Summary of the Number of Event
and Censored Values*

Total	Event	Censored	Percent Censored
2188	317	1871	85.51

*Summary of the Weighted Number of Event
and Censored Values*

Total	Event	Censored	Percent Censored
78174935	8861235	69313701	88.66

Variance Estimation

Method	Taylor Series
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Model Fit Statistics

Criterion	Without Covariates	With Covariates
-2 LOG L	317206280	301491570
AIC	317206280	301491602

Testing Global Null Hypothesis: BETA=0

Test	Test Statistic	Num DF	Den DF	p-Value
Likelihood Ratio (Unadj.)	15714710.6	16	Infty	<.0001
Likelihood Ratio (Adj.)	111.6310	4.674	Infty	<.0001
Wald	.	16	.	.

Warning: The model has more parameters than there are independent variance units. Variance estimates may be unstable.

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<i>Analysis of Maximum Likelihood Estimates</i>						
<i>Parameter</i>	<i>DF</i>	<i>Estimate</i>	<i>Standard Error</i>	<i>t Value</i>	<i>Pr > t </i>	<i>Hazard Ratio</i>
<i>sbp_avg</i>	14	0.006116	0.002455	2.49	0.0259	1.006
<i>LBXTC</i>	14	0.001071	0.002014	0.53	0.6034	1.001
<i>LBDHDL</i>	14	-0.000814	0.005787	-0.14	0.8902	0.999
<i>smoke_status Current</i>	14	0.763276	0.123957	6.16	<.0001	2.145
<i>smoke_status Former</i>	14	-0.133440	0.183139	-0.73	0.4782	0.875
<i>bmi_cat Obese</i>	14	0.530286	0.119839	4.42	0.0006	1.699
<i>bmi_cat Overweight</i>	14	0.165549	0.250837	0.66	0.5200	1.180
<i>bmi_cat Underweight</i>	14	0.710205	0.682122	1.04	0.3154	2.034
<i>diabetes 1</i>	14	0.720909	0.166061	4.34	0.0007	2.056
<i>RIDAGEYR</i>	14	0.117625	0.008021	14.66	<.0001	1.125
<i>RIAGENDR 2</i>	14	-0.536305	0.245604	-2.18	0.0465	0.585
<i>RIDRETH1 1</i>	14	-0.611696	0.158676	-3.85	0.0018	0.542
<i>RIDRETH1 2</i>	14	-0.839592	0.261217	-3.21	0.0062	0.432
<i>RIDRETH1 4</i>	14	0.213928	0.167949	1.27	0.2235	1.239
<i>RIDRETH1 5</i>	14	-1.483458	0.564523	-2.63	0.0199	0.227
<i>DMDEDUC2</i>	14	-0.165214	0.073257	-2.26	0.0406	0.848